Matthias Takele

Software Engineer

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EXPERIENCE

Guild
May 2025 – Present

Software Engineer II

- Launched a pharmacy education benefit serving **1,000+ learners**, building conditional React/TypeScript enrollment forms and Rails integrations supporting nearly **$18M** in approved benefits.
- Fixed duplicate-funding race conditions in Rails/PostgreSQL with idempotency safeguards and recovery from audit history, covering **80K+ approvals** totaling **$160M+**.
- Built a resumable Node.js migration pipeline to move legacy reimbursement data into new payments services, integrating PostgreSQL, DynamoDB, Snowflake, and GraphQL.
- Added configurable payouts for books, materials, and fees to a separate education benefits platform, extending Python/FastAPI calculations and React/TypeScript policy controls.

Datadog
Aug 2023 – May 2025

Software Engineer I

- Built Python ETL pipelines from Amazon S3 to PostgreSQL, validating incoming data and reducing query latency by **20%**.
- Developed a Go migration layer to move usage data onto a contract-aware schema while preserving existing customer-facing API behavior.
- Automated dataset deprecation and cleanup with scheduled jobs and monitoring, reducing stale data by **40%** and decreasing manual maintenance.

Datadog
Jun 2022 – Aug 2022

Software Engineering Intern

- Built a PostgreSQL migration tracker that caught schema discrepancies in containerized environments, reducing migration errors by **90%**.
- Aggregated and validated approximately **200M** historical records for usage and billing graphs, improving retrieval times by **30%**.

PROJECTS

Albie albie-app.com
2025 – Present

React Native/Expo · TypeScript · Node.js · PostgreSQL · Redis · Cloudflare R2 · Socket.IO

- Built and shipped an iOS app with **100+ beta users** for collaborative photo albums, journaling, and messaging using React Native/Expo, TypeScript, and a Node.js/PostgreSQL backend.
- Added semantic photo search and an on-device AI assistant using image-text embeddings, local LLMs, structured outputs, and response validation.
- Implemented resumable media uploads and offline synchronization with durable queues and idempotent APIs, using Redis, Cloudflare R2, and Socket.IO for shared albums.

RecipeVerse Demo · GitHub
Feb 2025

React · TypeScript · Firebase

- Built a recipe discovery app with React/TypeScript, TF-IDF search, CSV indexing, and Firebase authentication for saved recipes.

Spotify Top Artists Trivia Demo · GitHub
Dec 2024 – Jan 2025

Python · Flask · Jinja · Spotify Web API · Vercel

- Turned Spotify top artists into personalized trivia using the Spotify Web API, Flask, and Jinja templates; deployed on Vercel.

TECHNICAL SKILLS

Languages	Python, TypeScript, Go, Ruby, SQL
Frameworks	React, React Native/Expo, Rails, FastAPI, Node.js, Express, GraphQL
Data & Infrastructure	PostgreSQL, Redis, Snowflake, DynamoDB, AWS, Docker, Kubernetes

EDUCATION

Massachusetts Institute of Technology
May 2023

B.S. in Computer Science and Engineering